

Modeling Recidivism Rate to Determine Key Predictors of Reoffending

Wasim Khan, Ali Syed, Shrey Kothari, Usman Fawaz

University of California, Department of Statistics

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Professor Xinping Cui

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Introduction:

Identifying the leading causes as to why prison inmates can often become stuck in a pattern of returning to the justice system is something that has puzzled scientists and statisticians alike. Often studies, such as the Correctional Offender Management Profiling for Alternative Sanctions system, better known as the COMPAS system, are an example of how we choose to identify leading causes of why prisoners may reoffend, better known as recidivism rate. This study, compiled together with data, inmate information, and predictions, offers some kind of metric as to why some may turn back to the prison system rather than escape it.

Our project aims to investigate what specific aspects of this data, whether that be race, gender, juvenile misdemeanors, or just plainly aggression score, can be used to generalize or understand what specific demographics tend to end up entrapped by the prison system. Identifying such patterns, significant predictors, and explanations as to the complexity of the prison system may offer better solutions as to how we are meant to combat the growing cycle of prison reliance.

The research questions we aimed to answer within this follow step by step, meaning without one, it would diminish the importance of the following question. Given this, our main question of interest is what significant predictors play a role in increasing, or decreasing the likelihood of recidivism. Such predictors to name are race, gender, sex, juvenile felony rates, and an aggression score measured on a scale of one to ten, called the decile score. To understand the motivation as to why we want to isolate these metrics, specifically race and gender, it is important to identify the scrutiny that the COMPAS metric receives, often coming under fire due to racial bias or ethical malpractice. By selecting such predictors, we are not only opening up a conversation of if these stereotypes are correct, but we are also identifying if such predictors

have a genuine deviancy in how certain inmates are targeted within the prison system. This as a whole sets our initial model up for analysis, cleaning predictors not relevant to this hypothesis, fitting a linear model, and conducting diagnostic tests to determine if we have found the confirmation that our predictors are significant.

Following this, we planned to test more complex interactions between not only other predictors, but the predictors we have in front of us. As mentioned in our initial model, we purposefully included predictors such as decile score and juvenile felonies, not to reinforce our existing theory of racial bias, but to determine if such variables are directly correlated. Our reasoning for this is that sometimes isolating variables by themselves offers an indication as to how these variables suddenly become more or less significant when they are allowed to be directly tested with each other. Such an example for our second order model is the interaction between juvenile felony rates with race, testing if they influence recidivism. Such relationships being tested may lead us to significant results not shown by our initial model.

After we tested and ran all the models, to test accuracy and means of improvement, we used R^2 and RMSE measurements to identify how much variance is being explained by our model. Hence if there is an increased score, for example 13.56% without interaction terms and 15.54% with such interaction terms, this is one of the clearer signs that this interaction term may help improve our model, and furthermore identify that these predictors contribute to the recidivism rate. Using RMSE was also quite helpful for error metrics, measuring any error in predictions measuring our response variance, recidivism rate. Lastly using diagnostic plots like Q-Q plots and residual vs fitted plots helped us identify if our models are meeting certain criteria to be accurate models, which in this case is constant variance and linearity, and for the model with the interaction terms, we use partial F-tests to check if they are significant predictors.

Regression Analysis:

Our analysis was divided into four steps, each achieving an intended purpose of further exploring ways to accurately refine the model for accurately predicting the recidivism metric. Split into initial exploratory data analysis, initial model building, refined model building, and lastly identifying what was needed in our final model. Each model builds off of the previous, each time identifying significant predictors, measuring accuracy by analysis, and along the track doing summary statistics to ensure nothing limited the accuracy of our model.

Our first step of exploratory data analysis aimed to get some preliminary assumptions, not as to determine correlation but rather get a generalized idea as to how the data was spread and split up. For example, identifying that the prison system was majority african-american and caucasian, was quite important when considering the theory of racial bias. Another statistic worth mentioning would be the fact that most prison inmates do not have existing felonies as juveniles, meaning they are offenders after they turn adults. Why this was important is that we cannot simply use age as a metric, and we see this later on in our model significance results.

Our first order model was fit using the predictors; race, juvenile count, prior offenses, and decile score. This proved to be a highly effective initial model given it got an R^2 of 0.155, which roughly explains for around one seventh of the models variance, showing not a bad nor a good model in terms of the predictors explaining variability. What we will see later on is that this metric is not fully indicative of the correct variability, as it does decrease with later refinements. One other point of notice is that our RMSE score was shown to be 0.54, and although insignificant upon first glance, this is quite unreliable given our recidivism operates on a binary classification method, in which 1 signifies that the inmate reoffended in the two years from being

initially released, and 0 being that they did not. Some inmates were given -1, and this indicates that they were simply not accounted for. In this context, 0.54 would mean that there is a half and half chance that the model gets the prediction wrong, and this would not be good to have.

Our second model, considered our more refined model, built upon our initial model by adding an interaction term between juvenile felony count and race. This was intended to capture more effects of how race played into childhood felonies and if that played a factor in long term reliance on the prison system, hence higher recidivism rate. This proved to be somewhat effective, in our metrics, specifically R^2 we see an increase with a score of 0.157, and the RMSE decreasing with a score of 0.53, which is a slight but effective improvement. To fully test the impact of the sole interaction term, rather than place the entire model's success on this, we employed a partial F-test that indicated that the interaction term was indeed effective.

Our final refined model took to achieving a quite robust and proactive approach, only adding what elements we needed and ridding the model of that which we did not need. In a step between we had added curvature and accounted for error, which was attempted to be mitigated by the fact that we added such terms. Unfortunately this not only reduced R^2 but also increased general error through RMSE. To counter this we settled on a final model of the elements: race, age, age^2 , priors_count, $priors_count^2$, decile_score, $decile_score^2$, which not only took into account our valuable terms but also took necessary precautions of curvature terms. The end result was a model that achieved an R^2 of 0.1557, which although identical to the prior model R^2 , this time we were shown to finally have a statistically significant model for the first time. This proved to be a right direction in finally identifying the root causes of recidivism.

Conclusion and Limitations:

After we analyzed the COMPAS data and determined that several of our predictors are statistically significant, we saw that our final model explained only 15 to 16 percent of the variance for recidivism outcomes. This means that our model would have limited usefulness in the real world, especially in high-stakes contexts.

Our process highlighted several important challenges. Since our response variable is categorical, we wouldn't use linear regression. This was supported by diagnostic plots showing non-normal residuals and heteroscedasticity. This was why most, if not all, of our assumptions were violated. Also, there were very few improvements in model performance after we cleaned the model and decided which predictors were statistically significant. A better approach would be to use a classification model, such as logistic regression, because these are built for categorical response variables, such as *is_recid*.

This study also showed us that the data had limitations. Even after adding relevant predictors influencing recidivism, we get a low R^2 value. This means that other factors are not captured by the model. On top of that, the violations reduce the model's overall reliability.

Overall, this project's main goal is to experience the challenges of modeling the data and the importance of selecting both the appropriate model and the appropriate predictors with necessary interactions and quadratics. Our analysis provides insight into the relationships between recidivism and various predictors while demonstrating the limitations of linear regression in this particular setting.

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